

The University of Nottingham

SCHOOL OF Computer Science

A LEVEL THREE MODULE, SPRING SEMESTER 2018-2019

Computer Vision

Time allowed TWO HOURS

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

Answer THREE Out of FOUR Questions

Each question carries 20 marks

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

ADDITIONAL MATERIAL: None

INFORMATION FOR INVIGILATORS: None

1. The following questions are pertaining to Features, Object Recognition, and Convolutional Neural Networks.

- a. Scale Invariant Feature Transform (SIFT) consists of four main components, i.e., scale-space extrema detection, keypoint localisation, orientation assignment, and keypoint descriptors.

- i. Define the terms "scale-space" and "scale-space extrema".

[2 Marks]

- ii. To locate a keypoint, the sum of squared differences (SSD) plays a central role. The formula to calculate SSD of a local image window is given as

$$E(u, v) = \sum_{(x, y) \in W} (I(x + u, y + v) - I(x, y))^2$$

where $I(x, y)$ is the image pixel value at coordinates (x, y) , and (u, v) the vector representing a small shift of image window W . Clearly, SSD varies with (u, v) . Explain conceptually how (u, v) is used, in relation to SSD, to determine a keypoint.

[2 Marks]

- iii. What is the purpose of orientation assignment of the local patch centred at a keypoint? Explain, with the example in Figure 1, the possible consequence of omitting the step of orientation assignment.

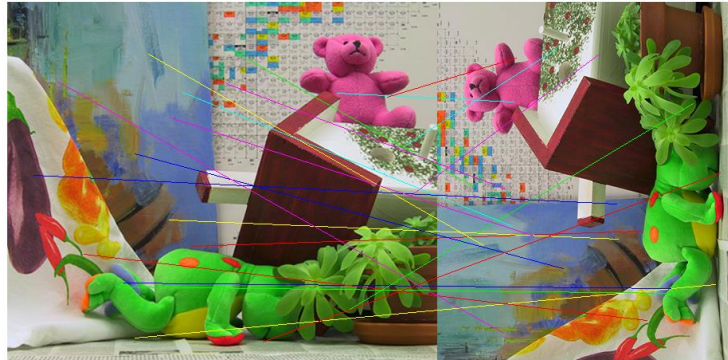


Image 1

Image 2

Figure 1

[2 Marks]

- iv. Describe the creation of the keypoint descriptor, including a justification of its main components.

[4 marks]

- b. Feature representation plays a vital role for object classification, e.g., to classify whether an image contains a cat or not.



Figure 2

- i. Briefly describe the concept of Bag of Features and illustrate the similarity and difference between the bag of features of image (a) and image (b) in Figure 2. [4 Marks]
- ii. Image features can also be captured by a Convolutional Neural Network (CNN) that is trained to classify the images. Explain how features can be learned by a CNN. [4 marks]
- iii. Assume that you have 100 samples of cat images and 100 samples of non-cat images. You can either use Bag of Features of these samples to train a linear Support Vector Machine (SVM), or use original images to train a CNN, to classify the images in Figure 2. Which method would you prefer to choose? Justify your answer. [2 Marks]

2. The following questions are pertaining to Stereo Vision, Motion, and Optical Flow.

- a. Disparity Gradient (DG) constraint has been exploited in several Stereo Vision algorithms to find the correspondence between pixels in left and right images.
 - i. Explain the terms "Disparity Gradient" and "Disparity Gradient constraint". [3 marks]
 - ii. Explain how the Disparity Gradient constraint was incorporated in the PMF algorithm. [3 marks]

- iii. In Zhang and Shan's progressive scheme, the Disparity Gradient constraint is translated to the disparity smoothness within a neighbourhood of pixels. Describe conceptually how pixels with known disparities within the neighbourhood of a pixel 'A' are used to determine the disparity uncertainty of the pixel 'A'.

[4 marks]

- b. Both Horn-Schunck and Lucas Kanade algorithms are designed for optical flow problems.

- i. Explain why Horn-Schunck is considered a global algorithm whilst Lucas Kanade a local algorithm.

[2 marks]

- ii. Describe what an image pyramid is. Using the Horn-Schunck algorithm as an example, explain how image pyramid can help speed up the optical flow algorithm.

[4 marks]

- c. Notice that both stereo vision and motion analysis involve directly and/or indirectly finding correspondences between two images, to obtain disparity for stereo vision problem and optical flow for motion analysis, respectively. Discuss whether it is possible to apply optical flow algorithms (e.g., Horn-Schunck or Lucas Kanade) to find disparity map for a pair of stereo images.

[4 marks]

3. The following questions are pertaining to Tracking and Segmentation.

- a. CONDENSATION is an effective algorithm for real tracking. It implements particle filters that are designed for multimodal tracking.

- i. Explain what multimodal tracking means.

[2 marks]

- ii. Describe what each particle represents in CONDENSATION algorithm.

[2 marks]

- iii. In CONDENSATION a particle is filtered through the process of tracking. In Figure 3 below, two particles are placed in frame (a), marked by 'A' and 'B', for tracking the person throwing discus. Assuming that at this point the tracker has been running for a while, particle 'A' is more likely to be filtered out than particle 'B' in (b) and (c). Explain why.

[6 marks]



Figure 3

- b. Active Contours are deformable models that can integrate segmentation and tracking for accurate object shape tracking. The modelling of Active Contour consists of internal energy and external energy.
- Describe the key components of the internal energy of an Active Contour. You do not have to use equations and can illustrate your description with sketches. [2 marks]
 - The external energy usually attracts the contour towards image features such as higher image gradients. Explain why such external energy is important. [2 marks]
 - An Active Contour is initialised on the image in Figure 4. An 8-neighbourhood search window is placed for each of the six key landmarks. Propose a possible set of positions of these six key landmarks and evaluate whether the set of positions would have higher or lower internal and external energy compared to the initial contour. You can illustrate your answer with sketches. [6 marks]

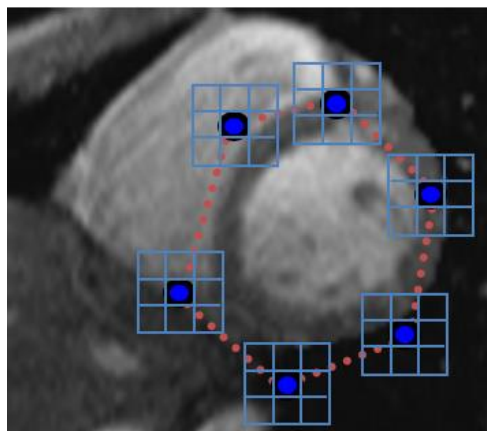


Figure 4

4. The following questions are pertaining to Object Recognition and Evaluation Methods.

- a. Viola-Jones object detection framework provides competitive object detection in real time. It was originally designed for face detection. It involves mainly three contributions including integral image calculation for fast feature extraction, boosting for feature selection, and attentional cascade for fast rejection of non-face windows.
- Explain what an integral image is and how it can be computed efficiently in one pass through the image. You can sketch to help your explanation. [3 marks]
 - Boosting is a type of Machine Learning algorithms that combines weak classifiers to form a strong classifier. Explain how boosting is used to select relevant features in Viola-Jones object detection. [6 marks]
 - To evaluate the performance of the Viola-Jones method for face detection, one can make use of precision and recall. Explain what precision and recall means and how they can be used to build Receiver Operating Characteristics (ROC) curves. [5 marks]
- b. Ground truth is often a popular reference for the evaluation of stereo vision methods. Explain what ground truth means and give one example of the ground truth for stereo vision problem. [2 marks]
- c. To evaluate how well an optical flow algorithm works, one can apply the optical flow obtained to the image at time t to predict the image at time $t + 1$, which can then be compared with the real image at time $t + 1$. Given a 4-by-4 image patch with intensity values at time t in Figure 5(a) and their corresponding optical flow vectors in Figure 5(b), sketch the prediction of the 4-by-4 image patch at time $t + 1$. Explain your sketch. [4 marks]

50	60	65	50
52	80	81	80
82	81	81	81
81	80	80	50

(a)

(0,0)	(0,0)	(0,0)	(0,0)
(0,0)	(1,1)	(1,1)	(1,1)
(1,0)	(1,1)	(1,1)	(1,0)
(1,1)	(1,2)	(1,2)	(0,0)

(b)

Figure 5